

Pneumococcal vaccines: A dose of change. Vaccine failures and breakthrough infections in Belgian children during the PCV13 era (2019–2024)

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ABSTRACT

Background: Since 2019, PCV13 has replaced PCV10 in Belgium's childhood vaccination program, reducing IPD cases caused by PCV13 serotypes, mainly 19A, but accompanied by serotype replacement. This changing epidemiology prompted consideration of the higher-valent PCV20 vaccine. Despite broader serotype coverage, clinical effectiveness against certain serotypes remains uncertain, raising concerns about vaccine failures. To date, vaccine failures and breakthrough infections have not been systematically studied in Belgium. This study aims to establish a baseline by characterizing vaccine failures and breakthrough infections in children 0–5 years old during the recent PCV13 era, providing a reference point ahead of PCV20 introduction.

Methods: We conducted a retrospective cohort study including children ≤5 years diagnosed with IPD in Belgium from 2019 to 2024, using surveillance data from the National Reference Centre for invasive pneumococci and the PediSurv network. Vaccine failures and breakthrough infections were identified and characterized by serotype and clinical characteristics.

Results: Of 543 pediatric IPD cases with known vaccination status, 451 (83 %) had received at least one dose of pneumococcal conjugate vaccine. Among these, we identified 26 vaccine failures (5.8 %) and 37 breakthrough infections (8.2 %). Differences in breakthrough proportions between PCV10 and PCV13 were primarily driven by PCV13-exclusive serotypes, after adjustment no significant differences remained. Vaccine failures were caused by serotypes 19A (61.5 %), 3 (26.9 %), and 14 (11.5 %), and were associated with older age (OR: 1.49; 95 % CI: 1.18–1.89) and invasive pneumonia (OR: 3.89; 95 % CI: 1.66–9.12). Breakthrough infections involved serotypes 19A (70.3 %), 3 (13.5 %), 14 (13.5 %), and 19F (2.7 %) and occurred more often in younger children (OR: 0.61; 95 % CI: 0.42–0.87).

Conclusion: These findings provide the first national baseline of vaccine failures, offering a reference point for evaluating PCV20 introduction. Continued monitoring will be essential to assess the real-world impact of higher-valent pneumococcal vaccines and to guide evidence-based vaccination policy.

1. Background

Streptococcus pneumoniae remains one of the leading causes of disease burden in young children worldwide, despite the availability of effective conjugate vaccines [1]. In Belgium, pneumococcal vaccination was introduced in the national childhood vaccination schedule in 2007 and the vaccine has changed over time from the 7-valent pneumococcal conjugate vaccine (PCV7) (Prevenar®; 2007–2011) to PCV13

(Prevenar13®; 2011–2016) and to the lower-valent PCV10 (Synflorix®; 2016–2019). In 2019, PCV13 was reintroduced in the national childhood immunization program, following a rise in invasive pneumococcal disease (IPD) in children with serotype 19A, a PCV13-exclusive serotype [2], and has been used ever since. The vaccine is administered following a 2 + 1 schedule at 2, 4 and 12 months of age, which has remained unchanged across all PCV formulations, and is available for all children free of charge. Vaccine uptake is high, with national pneumococcal

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vaccination coverage consistently estimated at 94 % for the complete schedule since 2015 [3].

After reintroduction of PCV13, the epidemiology of IPD in Belgium changed markedly, with a reduction in cases caused by PCV13 serotypes (mainly 19A) [4]. Simultaneously, an increase in non-PCV13 serotypes has been observed, a phenomenon known as serotype replacement. Notably, a substantial proportion of these non-PCV13 serotypes are included in the newer and higher-valent PCV20 vaccine. More recently, an exceptionally high IPD incidence was reported in Belgium during the 2024–2025 winter season, largely driven by vaccine serotypes [5]. In response to this shifting serotype distribution, and the overall rise in IPD incidence, Belgium's National Immunization Technical Advisory Group (NITAG) has recently recommended broadening protection by adopting the PCV20 vaccine for children in a 3 + 1 schedule [6]. This recommendation has not yet been implemented, and to date PCV13 remains the vaccine in place.

While PCV20 offers broader serotype coverage, recent studies suggest slightly lower immunogenicity for some shared serotypes compared to lower-valent vaccines [7,8]. Although PCV20 meets established non-inferiority criteria [9], its clinical effectiveness for certain serotypes remains uncertain due to the absence of a definitive correlate of protection. Lower immunogenicity could lead to an increase in vaccine failures and breakthrough infections. These are IPD cases caused by vaccine serotypes (VST) that occur in fully and partially vaccinated children, respectively [10]. Although pneumococcal conjugate vaccines offer substantial protection against IPD (vaccine efficacy: 88.7 % against PCV13 serotypes [11]), such cases can still occur. However, to date vaccine failure and breakthrough cases have never been systematically studied using Belgian national surveillance data.

In view of a potential change in the Belgian vaccination program, this study aims to establish a baseline by identifying and characterizing vaccine failures and breakthrough IPD cases in Belgian children aged 0–5 years during the most recent PCV13 era (2019–2024). We describe their frequency and serotype distribution, taking into account the relative occurrence of serotypes in our study population, and characterize associated patient characteristics. These findings will provide essential baseline data for evaluating the real-world impact of future PCV20 implementation and will support evidence-based IPD prevention and surveillance strategies in Belgium and other countries considering a transition to PCV20.

2. Methods

We conducted a retrospective cohort study in children ≤5 years old who were diagnosed with IPD between January 1, 2019 and December 31, 2024 in Belgium. IPD diagnosis was defined according to the European Center of Disease Control (ECDC) criteria as the isolation of *Streptococcus pneumoniae*, or detection of its nucleic acid or antigen, from a normally sterile site [12].

In Belgium, invasive *S. pneumoniae* isolates are sent to the National Reference Centre (NRC) at UZ Leuven, where capsular typing is performed. The NRC is estimated to receive isolates from about 95 % of IPD cases in Belgium [4]. For all received strains, *S. pneumoniae* identification was confirmed and serotyping was performed using the Quellung reaction with serotype-specific antisera (SSI Diagnostica, Denmark). Vaccination status, including vaccine type and number of doses, is reported on the laboratory request form submitted to the NRC. However, as this is a voluntary laboratory-based surveillance system, vaccination data from medical records are not routinely verified or completed by the laboratories sending isolates. Consequently, vaccination information was initially available for only 43 % of IPD cases (448/1047). To improve completeness, NRC data was linked with PediSurv, a voluntary pediatric sentinel surveillance network that collects additional vaccination information for IPD cases [13]. Although case numbers in PediSurv are low, vaccination status is recorded for more than 85 % of cases. Since PediSurv data are pseudonymized, cases were matched using

probabilistic matching based on date of diagnosis, age group, sex, and postal code. This resulted in known vaccination status for 52 % of NRC cases (543/1047), who were included in the final study population (Fig. 1).

Vaccination status was categorized as fully vaccinated (completion of the recommended 2 + 1 schedule), partially vaccinated (one or two doses), or unvaccinated. Partially vaccinated cases included two subgroups: individuals vaccinated according to age, meaning those who are following the vaccination schedule correctly but are too young to have received the final dose; and age-incorrectly vaccinated individuals, who have received fewer doses than recommended for their age.

Vaccine failures were defined as IPD caused by a vaccine serotype (VST) included in PCV10 or PCV13, occurring after full vaccination with the respective vaccine. Breakthrough infections were defined as IPD caused by vaccine serotypes in partially vaccinated children. Their proportions were calculated as the proportion of cases caused by vaccine serotypes among all vaccinated IPD cases aged ≤5 years, consistent with the definition commonly used in the literature, including the most recent systematic review [14] to allow international comparability. To assess potential differences by vaccine type, we stratified failures and breakthroughs by vaccine type (PCV10 or PCV13). To account for differences in serotype coverage, we repeated the comparison restricted to the ten serotypes included in both vaccines. Statistical significance was assessed using Fisher's exact test. Only children with a known and exclusive PCV10 or PCV13 vaccination schedule were included in the stratified analyses. Children with unknown, mixed, or other vaccine types were excluded from these subgroup comparisons, as precise attribution to a specific vaccine product was not possible.

Additionally, to compare the occurrence of failures and breakthroughs between serotypes, we calculated for each serotype the proportion of failures or breakthrough infections among all IPD cases caused by that serotype in our study population (regardless of vaccination status). This accounts for differences in the underlying frequency of each serotype and allows comparison of the relative susceptibility of serotypes to failures and breakthroughs across serotypes within our dataset.

To explore potential associations between vaccine failure and clinical characteristics, we performed univariate analyses comparing sex,

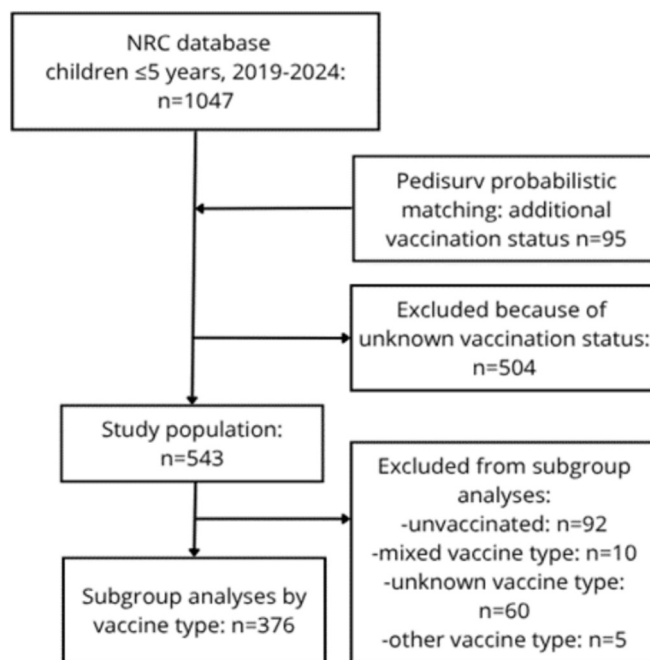


Fig. 1. Flowchart of case selection and subgrouping for restricted vaccine failure analyses.

age, clinical presentation, and outcome between vaccine failure or breakthrough cases and all other IPD cases without failure or breakthrough. For these analyses, cases were classified into three mutually exclusive categories: vaccine failures, breakthrough infections, and all other IPD cases. Each comparison compared one of the special groups (failures or breakthroughs) against the remainder of IPD cases, excluding the other special group. This avoided overlap between comparison groups and ensured that the reference population did not include either failures or breakthroughs.

It should be noted that the quality of outcome data is limited. As this is a laboratory-based surveillance system, patient outcomes are not always known to the reporting laboratories, and the timing of specimen submission to the NRC may affect death registration. Cases with missing outcome data were excluded from analyses involving the outcome variable. Furthermore, only deaths occurring within three days post-diagnosis are recorded, which may lead to an underestimation of true mortality. Despite these limitations, this variable was included to explore potential differences in severity between vaccine failure and non-failure cases.

Categorical variables were compared using odds ratios (OR) with 95 % confidence intervals (CI) and Fisher's exact test. Age was analyzed as a continuous variable using univariate logistic regression, reporting the OR per one-year increase. Because of anticipated limited case numbers, we planned to only perform multivariable analysis in case of associations in univariable analysis with $p < 0.2$. The same approach was applied to breakthrough infections among partially vaccinated children. Statistical significance was set at $p < 0.05$. All analyses were pre-specified at the start of the study based on the predefined study objectives and were performed using SAS Enterprise Guide 7.1 (SAS Institute, Cary, NC, USA).

3. Results

A total of 543 IPD cases in children aged ≤ 5 years with a known vaccination status were included in the study (Fig. 1). The majority (59 %) of the study population was male. Overall, regardless of vaccination status, serotypes 19A (18 %), 12F (12 %) and 24F (8 %) were the most common serotypes. There were 138 (25 %) IPD cases caused by PCV13 VSTs, mostly serotype 19A (70 % of all VSTs), followed by serotype 3 (20 %) and serotype 14 (7 %). Of the included children, 92 (17 %) were unvaccinated and 451 (83 %) had received at least one dose of a pneumococcal conjugate vaccine, of whom 207 (46 %) were fully vaccinated and 244 (54 %) partially vaccinated. Further descriptive characteristics of the study population can be found in Table 1. Vaccinated and unvaccinated children were comparable in terms of sex, age, clinical presentation, clinical outcome, and serotype group, with no significant differences in any of the variables (all $p > 0.05$; results not shown). The similar serotype distribution between vaccinated and unvaccinated children is consistent with high vaccination coverage in Belgium, which reduces circulation of vaccine serotypes even among unvaccinated children through herd immunity.

3.1. Vaccine failures and breakthrough infections

Among fully vaccinated children, vaccine failure occurred in 26 cases, one following PCV10 vaccination and 25 following PCV13 vaccination, corresponding to a failure proportion of 5.8 % (26/451) among all vaccinated IPD cases in children aged ≤ 5 years and 12.6 % (26/207) among fully vaccinated cases. Among partially vaccinated children, 37 breakthrough infections were observed, all occurring after PCV13 vaccination, corresponding to a breakthrough proportion of 8.2 % (37/451) among all vaccinated IPD cases and 15.2 % (37/244) among partially vaccinated cases.

When stratifying per vaccine type, vaccine failures were significantly more frequent among PCV13-vaccinated children (17.2 %) than PCV10-vaccinated children (2.4 %) (see Table 2). Similarly, breakthrough

Table 1

Baseline characteristics of the study population, stratified by vaccination status.

Characteristic	Total N (%) N = 543	Vaccinated N (%) N = 451	Unvaccinated N (%) N = 92
Age			
Mean (SD)	1.2 (1.45)	1.3 (1.41)	1.1 (1.59)
Median (IQR)	1 (1–2)	1 (0–2)	0 (0–2)
Sex			
Female	221 (40.7)	184 (40.8)	37 (40.2)
Male	322 (59.3)	267 (59.2)	55 (59.8)
Clinical presentation			
Bacteremia	230 (42.4)	189 (41.9)	41 (44.6)
Invasive Pneumonia ^a	209 (38.5)	177 (39.2)	32 (34.8)
Meningitis	79 (14.6)	63 (14.0)	16 (17.4)
Other	12 (2.2)	10 (1.8)	2 (2.2)
Outcome			
Alive	437 (80.5)	363 (80.4)	74 (80.4)
Deceased	9 (1.7)	7 (1.6)	2 (2.2)
Unknown outcome	97 (17.9)	81 (18.0)	16 (17.4)
Pneumococcal serotype^b			
PCV7	13 (2.4)	9 (2.0)	4 (4.3)
PCV10-not-7	1 (0.2)	0 (0)	1 (1.1)
PCV13-not-10	124 (22.8)	102 (22.6)	22 (23.9)
PCV15-not-13	45 (8.3)	36 (8.0)	9 (9.8)
PCV20-not-15	148 (27.3)	127 (28.2)	21 (22.8)
Non-vaccine type	212 (39.0)	177 (39.2)	35 (38.0)
Vaccine type (vaccinated only)			
PCV10	–	70 (15.5)	–
PCV13	–	306 (67.8)	–
Other/mixed/unknown	–	75 (16.6)	–
Vaccination status (vaccinated only)			
Fully vaccinated	–	207 (45.9)	–
Partially vaccinated, age-incorrectly	–	109 (24.2)	–
Partially vaccinated, age-appropriately	–	135 (29.9)	–

SD = standard deviation. IQR = interquartile range.

^a Invasive pneumonia is defined as pneumonia with bacteremia and/or empyema.

^b Serotypes were grouped based on inclusion in pneumococcal vaccines.

Table 2

Vaccine failure and breakthrough infections by vaccine type in children with exclusive PCV10 or PCV13 vaccination.

	Outcome	PCV10 n/N (%)	PCV13 n/N (%)	p-value
All VSTs	Vaccine failure	1/41 (2.4 %)	25/145 (17.2 %)	0.019
	Breakthrough	0/29 (0 %)	37/161 (23.0 %)	<0.001
Shared VSTs	Vaccine failure	1/41 (2.4 %)	3/145 (2.1 %)	0.528
	Breakthrough	0/29 (0 %)	6/161 (3.7 %)	0.593

VST = vaccine serotype. Bold values indicate statistically significant results ($p < 0.05$).

infections occurred only in the PCV13 group (23.0 %) while absent for PCV10. However, when restricting the analysis to serotypes included in both vaccines, no significant differences remained, indicating the observed differences were primarily driven by PCV13-exclusive serotypes (3 and 19A, as no cases with 6A occurred in Belgium during the study period).

3.2. Serotype distribution

Only three serotypes caused vaccine failures: primarily the PCV13-exclusive serotypes 19A (61.5 %) and serotype 3 (26.9 %), and less frequently the shared serotype 14 (11.5 %). Breakthrough infections involved the same serotypes 19A (70.3 %), 3 (13.5 %) and 14 (13.5 %), and one case due to shared serotype 19F (2.7 %). For comparison across serotypes, the proportion of vaccine failures and breakthrough infections among all IPD cases of each serotype were calculated, thereby accounting for differences in serotype frequency (Table 3). Serotype 14 had the highest proportion of failures with 30 %, compared to 26 % for serotype 3 and 17 % for 19A. Similarly, the proportion of breakthrough infections was highest for serotype 14 (50 %), followed by serotypes 19F (33 %), 19A (27 %) and 3 (19 %).

3.3. Characteristics of vaccine failures and breakthroughs

Demographic and clinical characteristics of vaccine failure and breakthrough cases compared to their respective control groups (all other IPD cases in study population, regardless of vaccination status) are presented in Table 4. Children with vaccine failure were significantly older, for each additional year of age, the odds of vaccine failure increased by 49 % (OR: 1.49; 95 % CI: 1.18–1.89). Vaccine failures had significantly higher odds of presenting with invasive pneumonia (OR: 3.89; 95 % CI: 1.66–9.12), and lower odds of bacteremia without pneumonia (OR: 0.31; 95 % CI: 0.11–0.83). No significant differences were observed between failure and non-failure cases with respect to meningitis, sex, or mortality. These associations remained statistically significant in multivariable models adjusted for age (Supplementary Table 1). Children with breakthrough infections were younger than non-breakthrough cases (OR: 0.61; 95 % CI: 0.42–0.87), but no other significant associations were observed.

4. Discussion

In our study of IPD cases 0–5 years old diagnosed between 2019 and 2024, we identified 26 vaccine failures and 37 breakthrough infections following respectively complete or partial vaccination with a pneumococcal conjugate vaccine, corresponding to failure and breakthrough proportions of 5.8 % and 8.2 % among all vaccinated IPD cases, respectively. These low proportions indicate vaccine failures and breakthroughs are uncommon, in line with the vaccines’ high effectiveness. Our findings are consistent with international studies. A 2022 systematic review of 26 studies reported an overall failure rate of 8.4 % and breakthrough rate of 9.3 %, though rates varied by vaccination schedule. Among studies using a 2 + 1 schedule, as in our study, the average failure rate was 4.2 % [14], while a recent Australian study reported 7.4 % [15].

Failures and breakthroughs in our study were limited to serotypes 19A, 3, 14, and a single breakthrough case involving serotype 19F. This pattern largely aligns with Mungall’s systematic review and other international studies, which consistently found serotypes 19A and 3 to be most commonly associated with failures and breakthroughs, followed by serotype 19F accounting for smaller proportions [14,16,17]. Other

PCV13 serotypes, including serotype 14, contributed minimally, with different serotypes sporadically reported in different countries, likely reflecting local epidemiology [14,15,18,19].

The predominance of these serotypes among vaccine failures and breakthroughs is consistent with the higher correlates of protection estimated for IPD caused by serotypes 3 (2.83 µg/mL), 19A (1.00 µg/mL), and 19F (1.17 µg/mL) compared to other vaccine serotypes, suggesting higher antibody concentrations are required to prevent IPD [20]. PCV13 effectiveness against serotype 3 has been shown to be low or absent in numerous studies, and lower effectiveness against 19A has also been reported [20–26]. These lower levels of protection may contribute to continued circulation of these serotypes, maintaining their relatively high prevalence in the population and, in turn, increasing the likelihood of vaccine failures and breakthroughs. In Belgium, serotypes 3 and 19A are particularly prevalent (together 90 % of all IPD by PCV13 VSTs), due to the country’s unique switch from PCV13 to PCV10 and back to PCV13. This high prevalence helps explain their predominance among observed vaccine failures and breakthroughs in our study.

Surprisingly, in our study, 11.5 % (*n* = 3) of vaccine failures and 13.5 % (*n* = 5) of breakthroughs were attributed to serotype 14, whereas Mungall et al. reported this serotype in fewer than 1 % of cases [14], and in no other studies has this serotype exceeded 5 % of breakthrough cases [15,18,27]. However, given the very small number of cases in our study, this difference should be interpreted with caution. In our dataset, we estimated that up to 80 % of IPD cases caused by serotype 14 occurred in vaccinated children (either vaccine failures or breakthrough infections). This is unexpected, as the reported correlate of protection for serotype 14 is only slightly higher than the aggregate 0.35 µg/mL threshold and vaccine effectiveness against this serotype has generally been reported to be high [11,20]. Half of serotype 14 breakthroughs and failures were registered in 2024. The changed epidemiology in Belgium, with an exceptional increase of serotype 14 IPD over all age groups in 2024 [5] may play a role.

Cases with vaccine failure were significantly older and breakthrough cases significantly younger. This reflects the vaccination schedule, since full vaccination is typically completed after the first year of life. In contrast, partially vaccinated cases still completing their schedule are per definition younger than one year old. Indeed 27 of 37 observed breakthroughs occurred in children under one, consistent with their vaccination status. After adjusting for age, vaccine failure cases remained more likely to present with invasive pneumonia and less likely with only bacteremia, compared to other IPD cases. This is consistent with findings from US and European studies linking failures to complicated pneumonia and empyema [18,19,28,29]. Although empyema could not be specifically assessed in our study, the association with pneumonia aligns with these findings.

This pattern is probably related to the predominance of serotypes 19A and 3 among failures, as these are known to be associated with complicated pneumonia and empyema [27,30,31]. Interestingly, breakthrough infections, though also dominated by 19A and 3, did not differ clinically from non-breakthrough cases. Factors beyond serotype distribution, such as individual host susceptibility, probably contribute as well. Unrecorded underlying risk factors (e.g., immunodeficiency, anatomical abnormalities) may predispose individuals to lung infections and have been linked with increased risk of vaccine failure [19,29]. However, as such data were not available in our study, this association could not be explored, highlighting an important opportunity for future research.

This study has several limitations. First, nearly half of the cases were excluded due to missing vaccination status. There is no clear reason to expect that vaccination status was more likely to be missing for vaccinated or unvaccinated children, and baseline characteristics (including age, sex, clinical presentation, and serotypes) were very similar between included and excluded cases, suggesting that any potential selection bias is probably minimal. Nevertheless, the substantial loss of data may have reduced the statistical power and precision of the findings. Possible

Table 3
Number of vaccine failures and breakthrough infections per serotype, and the proportion of failures and breakthroughs among all IPD cases of each serotype.

Serotype	Vaccine failures (N)	Breakthrough infection (N)	Total IPD cases (N)	Proportion of failures among cases	Proportion of breakthroughs among cases
19A	16	26	97	16.5 %	26.8 %
3	7	5	27	25.9 %	18.5 %
14	3	5	10	30.0 %	50.0 %
19F	0	1	3	0.0 %	33.3 %

Table 4

Clinical characteristics associated with vaccine failure and breakthrough infections.

	Failures (N = 26)	Breakthrough (N = 37)	Other IPD (N = 480)	Failures vs other IPD		Breakthroughs vs other IPD	
				OR (95 % CI)	p-value	OR (95 % CI)	p-value
Age (continuous)				1.49 (1.18–1.89)	<0.001	0.61 (0.42–0.87)	0.006
Sex, male	19	21	282	1.88 (0.77–4.55)	0.217	0.97 (0.49–1.92)	1.000
Invasive pneumonia	18	16	176	3.89 (1.66–9.12)	0.001	1.32 (0.67–2.59)	0.481
Bacteremia	5	16	209	0.31 (0.11–0.83)	0.015	0.99 (0.50–1.94)	1.000
Meningitis	2	3	74	0.46 (0.11–7.98)	0.424	0.48 (0.15–1.62)	0.337
Outcome, deceased	0	0	9	Not estimable	1.000	Not estimable	1.000

For descriptive comparisons, vaccine failure cases and breakthrough cases were each compared against all other IPD cases in the study population. Other IPD = all remaining IPD cases in the study population after exclusion of both failures and breakthroughs, regardless of vaccination status. OR = odds ratio; CI = confidence interval; p-values based on Fisher's exact test. Bold values indicate statistically significant results ($p < 0.05$).

misclassification of vaccination status could also have introduced information bias. Accurate vaccination data linkage with registries would improve robustness but is currently not feasible in Belgium. International comparisons and generalizability are limited by differences in IPD epidemiology, vaccine types, and immunization schedules. Despite these limitations, the study benefits from nationwide laboratory surveillance with high case coverage, providing a robust baseline prior to PCV20 introduction. To our knowledge, vaccine failures and breakthroughs have not been previously studied using Belgian data, and the use of real-world surveillance data ensures that these results can directly support evidence-based vaccination policies and future evaluations of PCV20.

Our findings are highly relevant for the upcoming PCV20 introduction. While PCV20 meets non-inferiority criteria, antibody levels (IgG GMCs) and opsonophagocytic activity (OPA GMTs) are generally slightly lower than those induced by PCV13, even with a 3 + 1 schedule [7]. Although the clinical significance of these modest reductions remains uncertain, they are concerning given the already lower effectiveness of PCV13 against certain serotypes. Optimizing vaccination policy will require balancing expanded serotype coverage with optimal protection against individual serotypes. These findings underscore the need for ongoing monitoring of vaccine effectiveness and potential vaccine failures following PCV20 implementation. To facilitate such monitoring, strengthening vaccination registries and ensuring complete, high-quality vaccination data should also be prioritized.

5. Conclusion

Vaccine failures and breakthrough infections in young children after PCV10 and PCV13 vaccination in Belgium are rare (5.6 % and 8.2 % of all vaccinated IPD cases, respectively) and limited to serotypes 19A, 3, 14, and 19F. This study provides the first national estimate of the frequency and serotype distribution of such cases, establishing an important baseline ahead of the planned introduction of the higher-valent PCV20 vaccine. Ongoing monitoring of vaccine failures post-PCV20 implementation will be crucial to evaluate its real-world impact, identify any gaps in effectiveness, and support evidence-based policy decisions to ensure optimal protection against severe invasive pneumococcal disease.

CRediT authorship contribution statement

Cato Dambre: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Laura Cornelissen:** Writing – review & editing. **Lize Cuypers:** Writing – review & editing, Data curation, Conceptualization. **Stefanie Desmet:** Writing – review & editing, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT (OpenAI) to assist with language editing and phrasing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the final version of the manuscript.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lize Cuypers reports a relationship with Merck & Co Inc that includes: travel reimbursement. Stefanie Desmet reports a relationship with Merck & Co Inc that includes: board membership and speaking and lecture fees. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2025.128065>.

Data availability

Data will be made available on request.

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